

Problem - Solution Fit Template

Phase 2: Project Design Phase

shopEZ E-commerce Marketplace | MERN Stack | June 2026

1. Problem - Solution Fit Matrix

This matrix maps each identified Customer Pain Point against the limitations of existing alternatives, and demonstrates how shopEZ's specific implementation resolves each problem.

Customer Pain Points	Existing Alternatives (Shopify, Monoliths)	shopEZ Solution Fit
Cart desync and item loss	Session-only cookies or basic local storage which clears on session timeout.	Persistent MERN Lifecycle: State is bound to React Context, syncing with local storage and database endpoints on login.
Poor catalog search and filtering	Simple string matches ignoring variants; return heavy, unfiltered datasets.	Dynamic Backend Filters: Mongoose queries support autocomplete suggestions, regex search, pricing boundaries, and rating filters.

Unoptimized media loading	Local server hosting bottlenecking bandwidth; high-resolution files served uncompressed.	Local Multer Storage + Static Serving: Uploaded images stored in /uploads with Express static serving for fast local access.
Complex administrative controls	Split admin pages or generic CMS dashboards with no real-time inventory alerts.	Unified Admin Ledger: Live analytics display monthly earnings graphs, user toggles, and low-stock indicators.
Checkout friction and hidden fees	Single-page checkout forms with no step guidance; totals revealed at last step.	Multi-Step Checkout Wizard: Address, Shipping, Coupon, Payment steps with itemized live totals (subtotal, discount, shipping, tax).
Payment trust and security	Redirect-based payment flows that break user trust and session state.	Mock Stripe Payment Intent: Client-side Stripe inputs via a simulated endpoint returning signed client secrets for secure transactions.
No order fulfillment visibility	Customers receive only a single "Shipped" email notification; no granular tracking.	6-Step Fulfillment Timeline: Visual stepper (Placed, Confirmed, Packed, Shipped, Out for Delivery, Delivered) updating in real-time from Admin panel.
Unverified product reviews	Any registered user can post reviews regardless of purchase history.	Verified Purchase Badge System: Review submission restricted to confirmed order holders; helpful upvote tallies maintain community trust.

2. Validation Summary

Key Finding: shopEZ's MERN stack implementation addresses all 8 primary pain points through native architectural advantages of the decoupled React + Express + MongoDB system, eliminating reliance on

external CMS platforms or plugin ecosystems.